

Q1: Genetic Algorithm

A robot must visit 5 checkpoints:

R1, R2, R3, R4, R5

Cost matrix:

	R1	R2	R3	R4	R5
R1	0	11	9	7	10
R2	11	0	6	12	8
R3	9	6	0	5	13
R4	7	12	5	0	4
R5	10	8	13	4	0

Parents:

- P1: [R3, R1, R5, R2, R4]
- P2: [R2, R4, R3, R5, R1]

a. One-Point Crossover (cut after position 3)

Generate children

b. Repair invalid chromosomes

c. Mutation

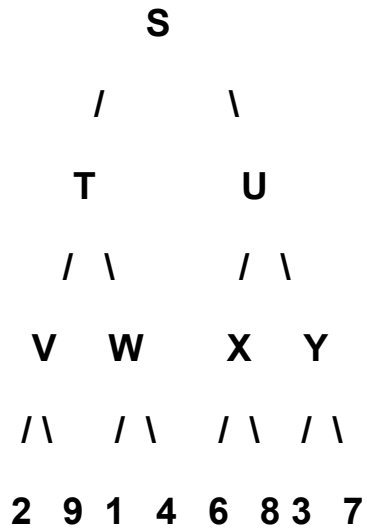
Apply scramble mutation on last 3 genes

d. Fitness Calculation

Compute path cost

Q2: Alpha Beta Pruning

Tree:



Tasks:

- Perform Minimax
- Perform Alpha-Beta Pruning
Indicate pruned nodes
- Compare:
 - Final value
 - Nodes evaluated vs pruned